

2016.0 RANGE ROVER (LG), 309-00

EXHAUST SYSTEM – TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the exhaust system, refer to the relevant description and operation section of the workshop manual. REFER to: (309-00D Exhaust System - TDV8 4.4L Diesel)

[Exhaust System](#) (Description and Operation),
[Diesel Particulate Filter](#) (Description and Operation),
[Diesel Particulate Filter](#) (Description and Operation).

INSPECTION AND VERIFICATION

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Diesel particulate filter ▪ Differential pressure sensor hoses ▪ Differential pressure sensor metal pipes and unions ▪ Differential pressure sensor mounting position	▪ Differential pressure sensor ▪ Differential pressure sensor connector ▪ Differential pressure sensor wiring harness

- integrity ▪ Metal fatigue ▪ Joints ▪ Mountings ▪ Clearance around components ▪ Damaged diesel particulate filter	▪ Electrical connector(s) ▪ Sensor(s) ▪ Engine control module
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3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step

4. If the cause is not visually evident, verify the symptom and refer to the symptom chart.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Amber warning message on instrument cluster - DPF FULL REFER TO HANDBOOK -	▪ Diesel particulate filter has more than 24.5g of soot & a poor drive cycle	▪ If AMBER DPF FULL REFER TO HANDBOOK message is displayed or customer advises of amber, with no other reported messages the customer needs to be advised that the message is advisory only and that their current drive cycle does not allow the system to clean the particulate filter. No repair is required, if the vehicle is driven on a highway AS DIRECTED IN THE HANDBOOK then the light will be extinguished and the system self healed, nothing more than this is required. The message is illuminated when the diesel particulate filter has more than 24.5g of soot & a poor drive cycle, but the vehicles current drive cycle is predominately urban, or engine temperature does not reach 50degC this inhibits diesel particulate filter regeneration ▪ Diesel particulate filter regeneration procedure is described at the bottom of this section

DTC Index

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00

NOTES:

- If the control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual (section B1.2), or determine if any prior approval programme is in operation, prior to the installation of a new module/component
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer approved diagnostic system)
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- If DTCs are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals

DTC	DESCRIPTION	POSSIBLE CAUSE	ACTION
P0030-11	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit short to ground	NOTE: - Circuit LPPH_A - ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was	▪ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to ground. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as

		expected ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to ground ■ Heated oxygen sensor failure	required
P0030-12	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit short to battery	NOTE: - Circuit LPPH_A - ■ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ■ Harness failure - Heated oxygen sensor heater control circuit short circuit to power ■ Heated oxygen sensor failure	■ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P0030-13	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Circuit open	NOTE: - Circuit LPPH_A - ■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input	■ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for open circuit. Repair harness as required. Clear DTC and retest

		in response to an output ▪ Harness failure - Heated oxygen sensor heater control circuit open circuit ▪ Heated oxygen sensor failure	▪ Check and install new heated oxygen sensor as required
P0030-4B	HO2S Heater Control Circuit (Bank 1, Sensor 1) - Over temperature	▪ The engine control module detected an internal temperature above the expected range ▪ Harness failure - Heated oxygen sensor heater control circuit short circuit to power ▪ Heated oxygen sensor failure	▪ Using the manufacturer approved diagnostic system check datalogger signals, Oxygen Sensor (O2S) Heater Duty Cycle Bank 1 Sensor 1 (0x03A1). Refer to the electrical circuit diagrams and check the heated oxygen sensor heater control (heater ground) circuit for short circuit to power. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P0130-11	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit short to ground	NOTE: - Circuit LPV_A - ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Harness failure - Heated oxygen sensor circuit short circuit to ground	▪ Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for short circuit to ground. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required

		Heated oxygen sensor component failure	
P0130-12	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit short to battery	NOTE: - Circuit LPV_A - - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected - Harness failure - Heated oxygen sensor circuit short circuit to power - Heated oxygen sensor component failure	- Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for short circuit to power. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required
P0130-13	O2 Sensor Circuit (Bank 1 Sensor 1) - Circuit open	NOTE: - Circuit LPPC_A - - LPV_A - - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Harness failure - Heated oxygen sensor circuit open	- Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuit. Repair harness as required. Clear DTC and retest - Check and install new heated oxygen sensor as required

		circuit ▪ Heated oxygen sensor component failure	
P0130-26	O2 Sensor Circuit (Bank 1 Sensor 1) - Signal rate of change below threshold	▪ The signal transitions more slowly than is reasonably allowed ▪ Exhaust system leak ▪ Fuel control system failure ▪ Heated oxygen sensor to engine control module circuit short circuit to ground, short circuit to power, high resistance ▪ Heated oxygen sensor failure	▪ Check for and rectify any exhaust leak between cylinder head and catalytic converter. Check heated oxygen sensor is correctly installed in exhaust manifold ▪ Check fuel control system for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor to engine control module circuit for short circuit to ground, short circuit to power, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P0133-00	O2 Sensor Circuit Slow Response (Bank 1 Sensor 1) - No sub type information	▪ Exhaust system leak ▪ Fuel control system failure ▪ Heated oxygen sensor to engine control module wiring shield high resistance ▪ Heated oxygen sensor failure	▪ Check for and rectify any exhaust leak between cylinder head and catalytic converter. Check heated oxygen sensor is correctly installed in exhaust manifold ▪ Check fuel control system for related DTCs and refer to the relevant DTC index ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor to engine control module wiring shield for high resistance. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required

P0426-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank 1, Sensor Circuit 1) - No sub type information	NOTE: - Circuit CCCIT_A - - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor pre catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required
P0427-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 1) - No sub type information	NOTE: - Circuit CCCIT_A - - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to ground - Exhaust gas temperature sensor pre catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required
P0428-00	Catalyst Temperature Sensor Circuit High (Bank 1, Sensor Circuit 1) - No sub type information	NOTE: - Circuit CCCIT_A - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to power, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required

		power, open circuit, high resistance Exhaust gas temperature sensor pre catalytic converter failure	
P042B-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank1, Sensor Circuit 2) - No sub type information	NOTE: - Circuit CCCOT_A - - Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground, open circuit, high resistance - Check and install new exhaust gas temperature sensor post catalytic converter as required
P042C-00	Catalyst Temperature Sensor Circuit Low (Bank 1, Sensor Circuit 2) - No sub type information	NOTE: - Circuit CCCOT_A - - Exhaust gas temperature sensor post catalytic converter short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground, open circuit, high resistance - Check and install new exhaust gas temperature sensor post catalytic converter as required
P042D-00	Catalyst Temperature Sensor Circuit High		Refer to the electrical circuit diagrams and check exhaust gas temperature sensor

	(Bank 1, Sensor Circuit 2) - No sub type information	NOTE: - Circuit CCCOT_A - - Exhaust gas temperature sensor post catalytic converter short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor post catalytic converter failure	post catalytic converter circuit for short circuit to power, open circuit, high resistance Check and install new exhaust gas temperature sensor post catalytic converter as required
P0435-00	Catalyst Temperature Sensor Circuit (Bank 2, Sensor Circuit 1) - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power - Exhaust gas temperature sensor post turbocharger circuit, short circuit to ground - Exhaust gas temperature sensor post turbocharger circuit, open circuit - Exhaust gas temperature sensor post turbocharger circuit, high resistance - Exhaust gas temperature sensor post turbocharger circuit, disconnected	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to power - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to ground - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for open circuit - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for high resistance - Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for disconnected - Check and install new exhaust gas temperature sensor post turbocharger as required

		Exhaust gas temperature sensor post turbocharger failure	
P0436-00	Catalyst Temperature Sensor Circuit Range/Performance (Bank 2, Sensor Circuit 1) - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power, short circuit to ground, open circuit - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to power, short circuit to ground, open circuit - Check and install new exhaust gas temperature sensor post turbocharger as required
P0436-16	Catalyst Temperature Sensor Circuit Range/Performance (Bank 2, Sensor Circuit 1) - Circuit voltage below threshold	NOTE: - Circuit STOT - - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas temperature sensor post turbocharger circuit, short circuit ground - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post turbocharger as required
P0437-00	Catalyst Temperature		Refer to the electrical circuit diagrams and check exhaust

	Sensor Circuit Low (Bank 2, Sensor Circuit 1) - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger circuit, short circuit to ground - Exhaust gas temperature sensor post turbocharger failure	gas temperature sensor post turbocharger circuit for short circuit to ground Check and install new exhaust gas temperature sensor post turbocharger required
P0438-00	Catalyst Temperature Sensor Circuit High (Bank 2, Sensor Circuit 1) - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger circuit, short circuit to power - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit, for short circuit to power - Check and install new exhaust gas temperature sensor post turbocharger as required
P0544-00	Exhaust Gas Temperature Sensor Circuit - Bank 1 Sensor 1 - No sub type information	NOTE: - Circuit CCCIT_A - - Exhaust gas temperature sensor pre catalytic converter circuit, short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor pre catalytic converter circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor pre catalytic converter as required

		pre catalytic converter failure	
P0547-00	Exhaust Gas Temperature Sensor Circuit - Bank 2 Sensor 1 - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger short circuit to ground - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required
P0548-00	Exhaust Gas Temperature Sensor Circuit Low - Bank 2 Sensor 1 - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger short circuit to ground - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required
P0549-00	Exhaust Gas Temperature Sensor Circuit High - Bank 2 Sensor 1 - No sub type information	NOTE: - Circuit STOT - Exhaust gas temperature sensor post turbocharger short circuit to power, open circuit, high resistance	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to power, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required

		Exhaust gas temperature sensor post turbocharger failure	
P2031-00	Exhaust Gas Temperature Sensor Circuit Bank 1 Sensor 2 - No sub type information	NOTE: - Circuit CCCOT_A - - Exhaust gas temperature sensor post catalytic converter short circuit to ground - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post catalytic converter as required
P2080-16	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 1 - Circuit voltage below threshold	NOTE: - Circuit CCCIT_A - - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Harness failure - Exhaust gas temperature sensor pre catalytic converter - Exhaust gas temperature sensor pre catalytic converter failure	- This DTC is set when there is a plausibility error on the signal from the exhaust gas temperature sensor pre catalytic converter. Refer to the workshop manual and check the exhaust gas temperature sensor pre catalytic converter and wiring harness for obvious signs of mechanical damage due to chaffing or heat. The exhaust gas temperature sensor pre catalytic converter is a thermistor with a signal and ground connection. Refer to the electrical circuit diagrams and check the signal circuit for open circuit, short circuit to power, short circuit to ground, high resistance. Check the ground circuit for open circuit, high resistance, short circuit to power. Repair the wiring harness as required - Check and install new

			exhaust gas temperature sensor pre catalytic converter as required
P2082-00	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 2 Sensor 1 - No sub type information	NOTE: - Circuit STOT - - Exhaust gas temperature sensor post turbocharger short circuit short circuit to ground, open circuit, high resistance - Exhaust gas temperature sensor post turbocharger failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post turbocharger circuit for short circuit to ground, open circuit, high resistance - Clear the DTC and retest - Check and install new exhaust gas temperature sensor post turbocharger as required
P2084-16	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 2 - Circuit voltage below threshold	NOTE: - Circuit CCCOT_A - - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Exhaust gas temperature sensor post catalytic converter circuit, short circuit to ground - Exhaust gas temperature sensor post catalytic converter failure	- Refer to the electrical circuit diagrams and check exhaust gas temperature sensor post catalytic converter circuit for short circuit to ground - Check and install new exhaust gas temperature sensor post catalytic converter as required
P2177-	System Too Lean	Oxygen	Check for excess fuel at

00	Off Idle - Bank 1 - No sub type information	concentration implausibly high ■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ■ Heated oxygen sensor failure	exhaust manifold, downpipe, heated oxygen sensor ■ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P2178-00	System Too Rich Off Idle - Bank 1 - No sub type information	■ Oxygen concentration implausibly low ■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ■ Heated oxygen sensor failure	■ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ■ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P2191-00	System Too Lean at Higher Load - Bank 1 - No sub type information	■ Oxygen concentration implausibly high ■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit	■ Check for excess fuel at exhaust manifold, downpipe, heated oxygen sensor ■ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest ■ Refer to the electrical circuit diagrams and check heated

		▪ Heated oxygen sensor failure	oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P2192-00	System Too Rich at Higher Load - Bank 1 - No sub type information	▪ Oxygen concentration implausibly low ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ▪ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P2195-00	O2 Sensor Signal Biassed/Stuck Lean - Bank 1, Sensor 1 - No sub type information	▪ Air leak at exhaust manifold, downpipe, heated oxygen sensor bank 1 ▪ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ▪ Heated oxygen sensor failure	▪ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest ▪ Check for air leaks at exhaust manifold, downpipe, heated oxygen sensor ▪ Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required

P2245-00	O2 Sensor Reference Voltage Circuit Low - Bank 1, Sensor 1 - No sub type information	NOTE: - Circuit LPV_A - ■ Harness failure - Heated oxygen sensor circuit ■ Oxygen sensor failure	■ This DTC is set if the engine control module detects the bank 1, heated oxygen sensor 1 reference voltage is lower than expected. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuits, high resistance, short circuit to ground. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P2246-00	O2 Sensor Reference Voltage Circuit High - Bank 1, Sensor 1 - No sub type information	NOTE: - Circuit LPV_A - ■ Harness failure - Heated oxygen sensor circuit ■ Oxygen sensor failure	■ This DTC is set if the engine control module detects the bank 1, heated oxygen sensor 1 reference voltage is greater than expected. Refer to the electrical circuit diagrams and check the heated oxygen sensor circuits for open circuits, high resistance, short circuit to power. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P2297-00	O2 Sensor Out of Range During Deceleration Bank 1, Sensor 1 - No sub type information	■ Heated oxygen sensor circuit short circuit to ground, short circuit to power, high resistance, open circuit ■ Heated oxygen sensor failure	■ Check for fuel / injector related DTCs and repair these first. Clear DTC and retest. Refer to the electrical circuit diagrams and check heated oxygen sensor for short circuit to ground, high resistance, open circuit. Repair harness as required. Clear DTC and retest ■ Check and install new heated oxygen sensor as required
P242A-	Exhaust Gas		■ Using the manufacturer

00	Temperature Sensor Circuit Bank 1 Sensor 3 - No sub type information	NOTE: - Circuit PFOT - ■ Harness failure - Exhaust gas temperature sensor post DPF ■ Exhaust gas temperature sensor post DPF failure	approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required ■ Check and install new exhaust gas temperature sensor post DPF as required
P242B-16	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 3 - Circuit voltage below threshold	NOTE: - Circuit PFOT - ■ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ■ Harness failure - Exhaust gas temperature sensor post DPF ■ exhaust gas temperature sensor post DPF failure	■ This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module ■ Refer to the workshop manual and check the exhaust gas temperature sensor post DPF for obvious signs of damage ■ Check the sensor harness for chaffing or heat damage ■ Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03F6), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03C8) ■ Refer to the electrical circuit diagrams and check the

			exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair wiring as required Check and install new exhaust gas temperature sensor post DPF as required
P242B-64	Exhaust Gas Temperature Sensor Circuit Range/Performance Bank 1 Sensor 3 - Signal plausibility failure	NOTE: - Circuit PFOT - - Harness failure - Exhaust gas temperature sensor post DPF - Exhaust gas temperature sensor post DPF failure	- Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas temperature sensor post DPF fails a cold start diagnostic check by the engine control module. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required - Check and install new exhaust gas temperature sensor post DPF as required
P242C-00	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 3 - No sub type information	NOTE: - Circuit PFOT - Harness failure - Exhaust gas	Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6). This DTC is set if the exhaust gas

		temperature sensor post DPF Exhaust gas temperature sensor post DPF failure	temperature sensor post DPF signal voltage is less than the engine control module was expecting. Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Repair harness as required Check and install new exhaust gas temperature sensor post DPF as required
P242D-17	Exhaust Gas Temperature Sensor Circuit High Bank 1 Sensor 3 - Circuit voltage above threshold	NOTE: - Circuit PFOT - - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Exhaust gas temperature sensor post DPF circuit, short circuit to power, open circuit - Exhaust gas temperature sensor post DPF failure	- Using the manufacturer approved diagnostic system check datalogger signals, Exhaust Gas Temperature Bank 1 Sensor 3 (0x03C8), Exhaust Gas Temperature Bank 1 Sensor 3 Voltage (0x03F6) - Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor post DPF circuit, for short circuit to power, open circuit - Check and install new exhaust gas temperature sensor post DPF as required
P244A-00	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - No sub type information	- Diagnostic failure check for minimum pressure differential characteristics - Diesel particulate filter internal components are	

		missing or destroyed	NOTE: If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) - Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index - Clear DTC and re-test - Check and install a new diesel particulate filter as required
P244A-96	Diesel Particulate Filter Differential Pressure Too Low (Bank 1) - Component internal failure	Destroyed particulate filter	NOTE: If this DTC is logged, refer to the relevant pinpoint tests in Section 309-00 (Exhaust System) Refer to the relevant pinpoint test in section 309-00
P244B-68	Diesel Particulate Filter Differential Pressure Too High (Bank 1) - Event information	- Engine protection back pressure high - Sudden increases in differential pressure across the diesel particulate filter	NOTE: This DTC when logged on its own is advisory only and no further action should be taken Using the manufacturer approved diagnostic system, check for related DTCs. If this DTC exists with any other diesel particulate filter differential pressure

			sensor DTCs, follow the advise given for that DTC
P2452-95	Diesel Particulate Filter Pressure Sensor A Circuit - Incorrect assembly	NOTE: In cold climates differential pressure sensor hose lines or metal pipes may be frozen ▪ Differential pressure sensor hoses connected incorrectly ▪ Differential pressure sensor hoses crushed, blocked, split	NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times,

			to allow a large negative offset value to adapt back to 0 Hpa - Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). Refer to the workshop manual and check differential pressure sensor hoses are installed correctly - Check differential pressure sensor hoses for crushed, blockage, split
P2453-00	Diesel Particulate Filter Pressure Sensor A Circuit Range/Performance - No sub type information	NOTE: - Circuit DPS - - Harness failure - Differential pressure sensor - Differential pressure sensor failure	NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved.

			Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa
			- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor fails a plausibility check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required - Check and install new differential pressure sensor as required
P2454-00	Diesel Particulate Filter Pressure		NOTE:

Sensor A Circuit
Low - No sub type
information

NOTE:

- Circuit DPS -

- Harness failure - Differential pressure sensor
- Differential pressure sensor crossed hose lines
- Differential pressure sensor failure

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored:

Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

			- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is less than the threshold set in the engine control module diagnostic check. Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to ground, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required - Check differential pressure sensor hose lines are installed correctly - Check and install new differential pressure sensor as required
P2455-00	Diesel Particulate Filter Pressure Sensor A Circuit High - No sub type information	NOTE: - Circuit DPS - - Harness failure - Differential pressure sensor - Differential pressure sensor failure - Exhaust back pressure is too high	NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate

filter pressure sensor
offset value to be
learnt and stored:
Using the
manufacturer
approved diagnostic
system, clear DTCs
from engine control
module, then monitor
the datalogger signal
'sump oil temperature
- measured' ensuring a
minimum of 50
degrees C is achieved.
Start engine, run
above 500RPM for 2
minutes, then a further
30 seconds at idle.
Ensure the engine
cooling fan is not
running. Set vehicle in
park and set ignition
status to off. Wait 30
seconds for the engine
control module to
power down, learn and
store diesel particulate
filter pressure sensor
offset value. This
process must be
carried out six times,
to allow a large
negative offset value
to adapt back to 0 Hpa

- Using the manufacturer approved diagnostic system, check engine control module, for related DTCs and refer to the relevant DTC index
- Using the manufacturer approved diagnostic system check datalogger signals, Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0x03DB). This DTC is set when the particulate pressure sensor voltage is greater than the threshold set in the engine control module diagnostic check.

			Refer to the electrical circuit diagrams and check the differential pressure sensor signal circuit for open circuit, short circuit to power, short circuit to other circuits. Check the sensor ground circuit for open circuit, short circuit to power, high resistance. Check the sensor power supply circuit for open circuit, short circuit to ground, high resistance. Repair harness as required ▪ Check and install new differential pressure sensor as required
P2456-00	Diesel Particulate Filter Pressure Sensor A Circuit Intermittent/Erratic - No sub type information	▪ Differential pressure sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ In cold climate or off road driving differential pressure sensor hoses crushed, blocked, split ▪ Differential pressure sensor failure	NOTE: If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored: Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved.

			Start engine, run above 500RPM for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa
			- Refer to the electrical circuit diagrams and check differential pressure sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check differential pressure sensor hoses for crushed, blocked, split - Check and replace diesel particulate filter differential pressure sensor as required
P2458-66	Diesel Particulate Filter Regeneration Duration (Bank 1) - Signal has too many transitions / events	Permanent regeneration	NOTE: This code is enabled for JLR engineering detailed diagnostics only. No further action should be taken

			Using the manufacturer approved diagnostic system, check for related DTCs and refer to the relevant DTC index
P2459-65	Diesel Particulate Filter Regeneration Frequency (Bank 1) - Signal has too few transitions / events	- Blocked regeneration - Customer driving routine does not allow the system to clean the particulate filter	NOTE: If DTC is P2459-65 or AMBER DPF FULL REFER TO HANDBOOK message is displayed with no other reported messages. No repair is required, if the vehicle is driven on a highway AS DIRECTED IN THE HANDBOOK then the light will be extinguished and the system self healed, nothing more than this is required - Refer to the diesel particulate filter regeneration procedure and carry out a diesel particulate filter regeneration - Advise customer of driving routine required to regenerate diesel particulate filter as stated in the vehicle handbook
P2463-00	Diesel Particulate Filter Restriction - Soot Accumulation (Bank 1) - No sub type information	Maximum soot mass	GO to Pinpoint Test A.
P2A00-16	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage below threshold	NOTE: - Circuit LPTR_A -	This DTC is set when the engine control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is less than the voltage threshold. This may be caused by the

		▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Harness failure - Heated oxygen sensor failure ▪ Heated oxygen sensor failure	heated oxygen sensor being too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all the heated oxygen sensor circuits for open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control module power and ground supplies. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as required
P2A00-17	O2 Circuit Range / Performance (Bank 1, Sensor 1) - Circuit voltage above threshold	NOTE: - Circuit LPTR_A - ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Harness failure - Heated oxygen sensor failure ▪ Heated oxygen sensor failure	▪ This DTC is set when the engine control module detects the voltage on the trim resistor circuit of the heated oxygen sensor is greater than the voltage threshold. This may be caused by the heated oxygen sensor being too hot to operate correctly. Refer to the workshop manual and check the exhaust system and heated oxygen sensor harness for sign of mechanical damage. Refer to the electrical circuit diagrams and check all the heated oxygen sensor circuits for open circuits, short circuit to power, short circuit to ground, short circuit to other circuits. Check all engine control module power and ground supplies. Repair harness as required. Clear DTC and retest ▪ Check and install new heated oxygen sensor as

Pre Catalyst Oxygen Sensor Adaption Process

Pre catalyst oxygen sensor adaption cycle is NOT possible while DTCs are logged. Using the manufacturer approved diagnostic system, carry out FIT NEW PARTICULATE FILTER PROCESS only and clear the DTCs. After completing FIT NEW PARTICULATE FILTER PROCESS continue with pre catalyst oxygen sensor adaption cycle as follows:-

1. Idle vehicle for 10 minutes
2. Set car in command shift 3rd gear
3. Accelerate to 3800RPM (where achievable) and overrun / coast down without braking until revs drop below 1500 RPM
4. Repeat step 3 a further 3 times
5. Check for any DTCs. If adaption has failed a DTC will be evident and the sensor will require replacement
6. Check oxygen sensor adaption is now complete by returning to pinpoint test A6 item 2
7. Using the manufacturer approved diagnostic system clear the DTC and re-test. Return vehicle to the customer

Differential Pressure Sensor Adaption Process

If a new diesel particulate filter pressure sensor or hose lines have been installed or incorrectly routed, or any pressure sensor circuit repairs carried out, the engine control module must learn and store the new diesel particulate filter pressure sensor offset value. The following conditions must be met to allow the diesel particulate filter pressure sensor offset value to be learnt and stored:

Using the manufacturer approved diagnostic system, clear DTCs from engine control module, then monitor the datalogger signal 'sump oil temperature - measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park and set ignition status to off. Wait 30 seconds for the engine control module to power down, learn and store diesel particulate filter pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 Hpa

PINPOINT TESTS

NOTE:

Using the results from steps 7 and 8 establish if the Soot Mass estimator and the differential pressure sensor are aligned, in terms of their assessment of soot in the filter. If the soot mass is less than 32.5 grams the diesel particulate filter is recoverable.

PINPOINT TEST A : RED WARNING MESSAGE DPF FULL VISIT DEALER IS DISPLAYED AND DTC P2463-00 IS LOGGED	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
A1: VIN INFORMATION	
	1 Record the full vehicle identification number for the vehicle under test, as this information may later be required by dealer technical support
	Has the vehicle identification number been recorded? Yes Proceed to the next step No Record the vehicle identification number and proceed to the next step
A2: VEHICLE MILEAGE	
	1 Record the full mileage for the vehicle under test, as this information may later be required by dealer technical support
	Has the mileage been recorded? Yes Proceed to the next step No Record the mileage and proceed to the next step
A3: RECORD ALL ENGINE CONTROL MODULE DTCS	
	1 Using the manufacturer approved diagnostic system, check engine control module, for DTCs
	Are any other DTCs logged? Yes Using the manufacturer approved diagnostic system refer to the relevant DTC index and repair all DTCs as requiredProceed to the next step

	No Proceed to step 7
A4: AMBER/RED DPF FULL MESSAGE	
	1 Verify with the customer the approximate time and mileage between the AMBER DPF FULL message being displayed and the RED DPF FULL message being displayed
	Was a mileage up to 625 miles (1000KM) driven between the AMBER DPF FULL message being displayed and the RED DPF FULL message being displayed? Yes Suspect the customer has ignored the AMBER DPF FULL REFER TO HANDBOOK message. The customer should be advised of this and the repair may become chargeable Proceed to step 7 No Proceed to the next step
A5: OTHER RELATED ENGINE CONTROL MODULE DTCS	
	1 Related engine control module DTCS other than P2463-00 are logged
	Are related engine control module DTCS other than P2463-00 logged? Yes Refer to the relevant DTC index. Repair as required. Using the manufacturer approved diagnostic system clear the DTCS and re-test No Proceed to step 7
A6: DIESEL PARTICULATE FILTER REGENERATION CYCLE	
	NOTE: Using the manufacturer approved diagnostic system begin to perform a service regeneration cycle, DO NOT drive the cleaning cycle. Record the grams of soot only 1 With the engine at running temperature check the pre catalyst oxygen sensor operation as follows:-
	2 Using the manufacturer approved diagnostic system check datalogger signals - Oxygen Sensor Voltage - (0xF424) ▪ Idle vehicle and record, should be 0.9 ~ 1 volt ▪ Idle to 100% pedal, during acceleration record, should be 0 ~ 0.3 volts ▪ Accelerate to 4500RPM (stationary) and close throttle, should read 1.1 ~ 1.3 volts
	Does the pre catalyst oxygen sensor meet this criteria?

	Yes Pre catalyst oxygen sensor is operating correctly Note: DO NOT carry out the adaption process on the pre catalyst oxygen sensor, as this resets the adaption to 0 Proceed to step 8 No Pre catalyst oxygen sensor adaption cycle is required Pre catalyst oxygen sensor adaption cycle is NOT possible while DTCs are logged Using the manufacturer approved diagnostic system, carry out FIT NEW PARTICULATE FILTER PROCESS only and clear the DTCs After completing FIT NEW PARTICULATE FILTER PROCESS continue with pre catalyst oxygen sensor adaption cycle as follows:- 1. Idle vehicle for 10 minutes 2. Set car in command shift 3rd gear 3. Accelerate to 3800RPM (where achievable) and overrun / coastdown without braking until revs drop below 1500 RPM 4. Repeat step 3 a further 3 times 5. Check for any DTCs, if adaption failed a DTC will be evident and the sensor will require replacement 6. Check oxygen sensor adaption is now complete by returning to B7 item 27. Using the manufacturer approved diagnostic system clear the DTC and re-test. Return vehicle to the customer
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A7: SOOT ESTIMATOR IS ACCURATE AND THE DIFFERENTIAL PRESSURE SENSOR READING IS WORKING CORRECTLY

	1 Using the manufacturer approved diagnostic system check datalogger signal - Particulate Filter Differential Pressure Sensor Voltage - Bank 1 - (0x03DB) - Record value with ignition ON - Idle engine record value - Hold engine speed at 4000RPM or maximum possible RPM and record value - Note: 1volt = 100mbar, 1.5volts = 300mbar at 4000RPM
	Have the values been recorded for each of the conditions? Yes Proceed to step 9 No Record values for each of the conditions and proceed to step 9

A8: EXHAUST GAS TEMPERATURES

	1 Using the manufacturer approved diagnostic system check datalogger signals - Exhaust Gas Temperature Bank 1 Sensor 2 - (0x03F5) - Exhaust Gas Temperature Bank 1 Sensor 1 - (0x03F4) - Exhaust Gas Temperature Bank 1 Sensor 2 - (0x03F5) - Exhaust Gas Temperature Bank 1 Sensor 3 - (0x03F6)
	2 Hold engine speed at 2000RPM for 2 minutes and record values (By doing this we are trying to establish if the system can recognize heat in the exhaust during regeneration)

	Are all of the exhaust gas temperature sensors showing reasonable values between 120°C and 400°C? Yes Proceed to step 10 No Check and install new exhaust gas temperature sensors as required. Refer to the new module/component installation note at the top of the DTC index Using the manufacturer approved diagnostic system clear the DTCs and re-test
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A9: COMPARISON OF SOOT MASS IN DIESEL PARTICULATE FILTER AND PARTICULATE FILTER DIFFERENTIAL PRESSURE SENSOR VOLTAGE

	1 Using the results from step 7 check soot mass is less than 32.5 grams
	2 Using the results from step 8 check particulate filter differential pressure sensor voltage at 4000RPM is less than 1 Volt (100mbar)
	Is diesel particulate filter soot mass value less than 32.5 grams and particulate filter differential pressure sensor voltage at 4000RPM less than 1 Volts (100mbar)? Yes If the diesel particulate filter soot mass value is greater than 31.5 grams and the differential pressure sensor voltage is between 0.5Volts and 0.9 Volt then the diesel particulate filter is low on soot but has not been driven to allow pressure correction of the diesel particulate filter, if other issue from the tests performed are evident. Proceed to next step No If the diesel particulate filter soot mass value is greater than 32.5 grams and the differential pressure sensor voltage greater than 1 Volt then the diesel particulate filter has a high soot contentIf no other issue from the tests performed are evident then the vehicle has a soot generated fault not detected by DTCs.Using the manufacturer approved smoke tester carry out a boost system leakage checkRemove the diesel particulate filter from the secondary turbo charger, lower the exhaust system and inspect for oil contaminationContact dealer technical support with all of the recorded values from the above tests

A10: SOOT MASS REDUCTION

	1 With the vehicle fully up to temperature and in Park maintain 2500RPM for 3 minutes
	2 Return to step 7, check diesel particulate filter soot mass value
	Has the diesel particulate filter soot mass reduced from the original reading? Yes Proceed to next step No Proceed to step 12

A11: DRIVE VEHICLE

	CAUTION: At all times during this procedure you should observe all relevant speed limits, laws, and regulations 1 Drive the vehicle until the engine reaches normal operating temperature. The engine should NOT be left idling to achieve working temperature. Drive the vehicle for a further twenty minutes, keeping the vehicle at a constant speed between 75 km/h (45 mph) and 120 km/h (75 mph). Keeping a constant speed enables the diesel particulate filter to regenerate more efficiently. It is therefore recommended that cruise control is used to achieve this, if possible (Do NOT carry out diesel particulate filter service regeneration)
	Is the diesel particulate filter soot mass less than 6 grams? Yes Using the manufacturer approved diagnostic system clear the DTC and re-test. Return vehicle to the customer No Contact dealer technical support with all of the recorded values from the above tests

A12: CARRY OUT FIT NEW PARTICULATE FILTER PROCESS

	WARNING: DO NOT carry out this process on any other occasion without first installing a new diesel particulate filter 1 Using the manufacturer approved diagnostic system, select SPECIAL APPLICATIONS ~ POWERTRAIN ~ carry out FIT NEW PARTICULATE FILTER PROCESS
	2 Carry out diesel particulate filter service regeneration. Record grams of soot following diesel particulate filter service regeneration
	Is the diesel particulate filter soot mass less than 6 grams? Yes Using the manufacturer approved diagnostic system clear the DTC and re-test. Return vehicle to the customer No Contact dealer technical support with all of the recorded values from the above tests

DIESEL PARTICULATE FILTER SOOT LEVEL INFORMATION

SOOT LEVEL SYMPTOM	VALUES
Soot level accumulation in diesel particulate filter is stopped	32.5g
Maximum soot level accumulation before regeneration process is inhibited	28.5g
RED warning lamp ON	31.5g
RED warning lamp OFF	24.5g
AMBER warning lamp ON	24.5g
AMBER warning lamp OFF	22.5g

DIESEL PARTICULATE FILTER REGENERATION PROCEDURE

If DPF FULL SEE HANDBOOK appears in the message center, carry out the following procedure

CAUTIONS:

- The regeneration procedure produces high temperatures in the diesel particulate filter. Heat can be felt radiating from beneath the vehicle, which is normal and not a cause for concern. However, the vehicle should not be parked over combustible material, particularly during dry weather. The heat generated could be sufficient to start a fire when in close proximity to combustible material such as long dry grass, paper etc
- At all times during this procedure you should observe all relevant speed limits, laws, and regulations

NOTES:

- The ideal speed and conditions for regeneration are 100 km/h (62 mph) ~ 120 km/h (75 mph), in Drive. Keeping a constant speed enables the diesel particulate filter to regenerate more efficiently. It is therefore recommended that cruise control is used to achieve this, if possible
- When driving off-road during the regeneration process, greater accelerator pedal use may be required

1. Drive the vehicle until the engine reaches normal operating temperature. The engine should **NOT** be left idling to achieve working temperature
2. Drive the vehicle for a further twenty minutes, keeping the vehicle at a constant speed between 75 km/h (45 mph) and 120 km/h (75 mph)
3. If regeneration is successful the warning message will be extinguished, once the message is extinguished please keep driving for 10 minutes to ensure that the diesel particulate filter is completely clean
4. If the message remains repeat the process

DIAGNOSTIC PROCEDURES TO BE USED
IF DTC P244A-00, P244A-95 OR P244A-
96 IS LOGGED IN THE ENGINE CONTROL
MODULE

PINPOINT TEST B : P244A-00, P244A-95 OR P244A-96	
TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
B1: PRE-CHECK (ENGINE CONTROL MODULE SOFTWARE)	
	1 Check the latest relevant level of software is installed to the engine control module

	Is the latest relevant level of software installed to the engine control module? Yes Proceed to the next step No 1. Using the manufacturer approved diagnostic system, re-configure the engine control module with the latest level software2. Proceed to the next step
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B2: PRE-CHECK (LOGGED DTCS)

	1 Check that there are not other engine control module DTCs logged
	Are other engine control module DTCs logged? Yes 1. Refer to the relevant DTC index and rectify as required2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the diagnosis menu tab3. Proceed to the next step No Proceed to the next step

B3: PRE CHECK (DIFFERENTIAL PRESSURE SENSOR ADAPTION)

	1 Check the differential pressure sensor offset value. The following conditions must be achieved to allow the differential pressure sensor offset value to be learnt - stored: Using the manufacturer approved diagnostic system, monitor the datalogger signal 'sump oil temperature measured' ensuring a minimum of 50 degrees C is achieved. Start engine, run above 500rpm for 2 minutes, then a further 30 seconds at idle. Ensure the engine cooling fan is not running. Set vehicle in park - set ignition status to off. Wait 30 seconds for the engine control module to power down, learn - store differential pressure sensor offset value. This process must be carried out six times, to allow a large negative offset value to adapt back to 0 hPa
	Has the differential pressure sensor adaption offset value check been carried out? Yes Proceed to the next step No 1. Carry out the differential pressure sensor adaption offset value check2. Proceed to the next step

B4: DIFFERENTIAL PRESSURE SENSOR, HOSE CHECKS

	1 Check the differential pressure sensor hoses for splits, crushing, connectivity, blockages or contamination
	Were the differential pressure sensor hoses split, crushed, blocked or contaminated? Yes 1. Rectify as required2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the

	diagnosis menu tab No Proceed to the next step
B5: DIFFERENTIAL PRESSURE SENSOR, CONNECTOR CHECKS	
	1 Check the differential pressure sensor connector for disconnected, connector pin is backed out, connector pin corrosion
	Was the differential pressure sensor connector disconnected, connector pins backed out, connector pin corroded Yes 1. Rectify as required 2. Clear the DTC and retest. When using the manufacturer approved diagnostic system clear all stored DTCs from the diagnosis menu tab No Proceed to the next step
B6: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT	
	NOTE: The following checks are designed to verify the differential pressure sensor, pressure/voltage transfer function and will determine if the differential pressure sensor operates within tolerance. The checks will also check for leakage within the differential pressure sensor assembly and pipework. The measured voltages of the differential pressure sensor will be the result of pressure being applied to the reference input and high pressure input on the differential pressure sensor.
	1 Connect a vehicle battery stabilizer
	2 Set the vehicle to park
	3 Set ignition to 'ON' engine 'OFF'
	4 Allow voltage to stabilize
	5 Disconnect both the rubber hoses from the metal pipes of the diesel particulate filter (the rubber hoses remain connected to the differential pressure sensor).
	6 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 0.49 volts & 0.51 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B7: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED ZERO)

	1 Install the manufacturer approved hand vacuum pump to the reference pipe and differential pressure sensor. Ensure the gauge is reading zero value for vacuum and pressure
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 0.485 volts & 0.515 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B8: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 5 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 5 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 1.35 volts & 1.47 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B9: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 10 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 10 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 2.30 volts & 2.45 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B10: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 15 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 15 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 3.30 volts & 3.45 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B11: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 20 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 20 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 4.25 volts & 4.40 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B12: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (VACUUM APPLIED 25 INHG)

	1 Using the manufacturer approved hand vacuum pump apply a vacuum of 25 inHg to the reference pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 4.75 volts & 4.90 volts Yes Release the vacuum and remove the hand vacuum pump from the reference pipe and differential pressure sensorProceed to the next step No Check and install a new differential pressure sensor as requiredCarry out the differential pressure sensor adaption offset value learn as described in test step B3

B13: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED ZERO)

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	1 Install the manufacturer approved hand vacuum pump to the high pressure pipe and differential pressure sensor. Ensure the gauge is reading zero value for vacuum and pressure
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 0.490 volts & 0.515 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B14: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 20KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 20 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 1.60 volts & 1.75 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B15: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 40KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 40 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 2.70 volts & 2.85 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B16: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 60KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 60 KPa to the high pressure pipe and differential pressure sensor

	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 3.80 volts & 3.90 volts Yes Proceed to the next step No Check and install a new differential pressure sensor as required

B17: DIFFERENTIAL PRESSURE SENSOR VOLTAGE MEASUREMENT (PRESSURE APPLIED 80KPA)

	1 Using the manufacturer approved hand vacuum pump apply a pressure of 80 KPa to the high pressure pipe and differential pressure sensor
	2 Using the manufacturer approved diagnostic system check datalogger signal – Particulate Filter Differential Pressure Sensor Voltage - Bank 1 (0X03DB) -
	Does the voltage measure between 4.0 volts & 4.89 volts Yes The differential pressure sensor is OK and should not be replaced Carry out the differential pressure sensor adaption offset value learn as described in test step B3 No Check and install a new differential pressure sensor as required Carry out the differential pressure sensor adaption offset value learn as described in test step B3

DTC INDEX

For a list of DTCs that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).